

Kavya Prakash

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Portfolio | LinkedIn

PROFESSIONAL SUMMARY

M.Sc. graduate in Biology (Immunology and Microbiology) from Uppsala University with experience in stem cell biology, epigenetics, neurodevelopmental toxicology, and transcriptomic data analysis. Skilled in qPCR, pyrosequencing, immunofluorescence, stem cell culture, RNA-seq analysis, and functional genomics. Seeking a PhD position in cell and molecular biology with a focus on disease mechanisms and stem cell-based models.

TECHNICAL SKILLS

Molecular Biology	qPCR, RT-qPCR, pyrosequencing (bisulfite-based DNA methylation), RNA extraction, DNA extraction, immunofluorescence staining
Cell Culture & Models	Human and mouse embryonic stem cells (hESC, mESC), SH-SY5Y neuroblastoma cells, Caco-2 intestinal epithelial cells, dorsal forebrain differentiation protocols
Bioinformatics	R (DESeq2, ggplot2), RNA-seq analysis, differential gene expression, Gene Ontology and KEGG enrichment analysis, PCA, heatmaps, Linux command line
Additional Techniques	ELISA, Western blotting, flow cytometry, whole-mount in situ hybridisation, live-cell imaging, GLP, QA/QC procedures
Laboratory Operations	Autoclaving, glassware washing, waste disposal, cell culture hood and equipment upkeep, consumables restocking, and supply ordering/vendor communication
Languages	English (IELTS 8.0), Malayalam (native), Swedish (basic)

EDUCATION

M.Sc. in Biology — Immunology and Microbiology <i>Uppsala University, Sweden</i>	<i>2024 – 2026</i>
M.Sc. in Biotechnology <i>Kerala University, India</i>	<i>2019 – 2021</i>
B.Sc. in Biotechnology <i>Kerala University, India</i>	<i>2016 – 2019</i>

SELECTED RESEARCH PROJECTS

Master's Thesis — Uppsala University <i>EpiTox Group, Department of Organismal Biology</i> <i>Cellular and Molecular Impact of BPA and BPF on Human and Mouse Embryonic Stem Cell Models of Dorsal Forebrain Differentiation</i>	<i>Oct 2025 – May 2026</i>
- Established and optimised a dorsal forebrain differentiation protocol in human and mouse embryonic stem cells to model early cortical neurodevelopment in vitro. - Characterised the effects of bisphenol A (BPA) and bisphenol F (BPF) on neuroectodermal differentiation markers and cellular stress responses using RT-qPCR and immunofluorescence staining. - Integrated molecular and imaging data to evaluate EDC-mediated disruption of epigenetic and transcriptional programmes during neural induction.	
M.Sc. Thesis — Kerala University <i>Kerala University, India</i> <i>Identification of microRNAs Regulating Differentially Expressed Genes in Cisplatin-Resistant Ovarian Cancer Cells</i>	<i>2019 – 2021</i>
- Profiled microRNA-mediated regulation of chemoresistance pathways in ovarian cancer through expression analysis and bioinformatics-assisted pathway interpretation. - Identified candidate miRNA-mRNA regulatory axes associated with platinum-based drug resistance mechanisms.	

BIOINFORMATICS PROJECT

Transcriptomic Analysis of Valproic Acid-Induced Neurodevelopmental Alterations in Human Forebrain Organoids *Independent Bioinformatics Research Project*

- Re-analysed publicly available RNA-seq data from human forebrain organoids exposed to valproic acid (VPA), a neurodevelopmental toxicant associated with autism spectrum disorder risk.
- Performed transcript quantification, differential gene expression analysis, Gene Ontology enrichment, and KEGG pathway analysis using Salmon, DESeq2, and R.
- Identified 7,151 significantly differentially expressed genes ($FDR < 0.05$), including 1,676 high-confidence DEGs with absolute $\log_2$ fold change greater than 1.
- Observed enrichment of pathways associated with ciliogenesis, neuronal development, synaptic signalling, myelination, and chromatin remodelling.
- Repository: <https://github.com/kavyaprakash-vks/vpa-forebrain-organoid-rnaseq>

RESEARCH EXPERIENCE

Research Trainee

Jul 2025 – Sep 2025

Uppsala University, Department of Organismal Biology — Physiology and Environmental Toxicology

- Differentiated SH-SY5Y neuroblastoma cells toward a glutamatergic neuronal phenotype and validated neuronal marker expression using qPCR.
- Maintained SH-SY5Y and Caco-2 cell cultures and optimised molecular protocols for toxicological studies.
- Managed laboratory maintenance and operations, including autoclaving, glassware washing, waste disposal, cell culture hood and equipment upkeep, consumables restocking, and supply ordering.

TEACHING EXPERIENCE

Laboratory Teaching Assistant

Nov 2025 – Jan 2026

Uppsala University

- Instructed undergraduate and graduate students in pyrosequencing, RT-qPCR, zebrafish whole-mount in situ hybridisation, Drosophila behavioural assays, and live-cell imaging.
- Supported laboratory operations, sample preparation, equipment setup, and experimental workflow design.

INDUSTRY EXPERIENCE

Quality Assurance & Quality Control Specialist

Oct 2021 – Jun 2024

Nexus Agro Farms, Kerala, India

- Implemented QA/QC procedures for hybrid crop testing and applied data-driven approaches to improve product consistency and regulatory compliance.

LEADERSHIP & COMMUNITY ENGAGEMENT

International Student Ambassador, Office of Science and Technology — Uppsala University *2025 – 2026*

CONFERENCE PRESENTATIONS

Kavya Prakash. “Environmental Pollutants and Brain Health: Insights from Human Cell Models of Neurodegeneration.” *Uppsala University Student Conference in Science and Technology*, 2025.

REFERENCES

Available upon request.